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Subject : Cloud Computing

**Q1. Explain the concept and use of Amazon Elastic Block Store (EBS).**

**Concept:**
Amazon **Elastic Block Store (EBS)** is a **block-level storage service** provided by **Amazon Web Services (AWS)**. It is designed to store **data persistently** and can be attached to **Amazon EC2 (Elastic Compute Cloud)** instances. Each EBS volume acts like a **hard drive** that you can use with your virtual server in the cloud.

EBS provides **high-performance, durable, and scalable** storage for workloads such as databases, applications, and operating systems. The data stored in an EBS volume remains available **even after the EC2 instance is stopped or terminated** (unless the volume is deleted manually).

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**Key Features of EBS:**

1. **Persistent Storage:**
   Data remains intact even after the EC2 instance is stopped or restarted.
2. **Scalability:**
   You can increase storage capacity and performance easily without downtime.
3. **High Availability & Durability:**
   Data is automatically replicated within its Availability Zone to protect against hardware failure.
4. **Snapshot Support:**
   EBS allows you to take **snapshots (backups)** of your volumes, which can be stored in Amazon S3 and used to restore data later.
5. **Performance Options:**
   Different volume types are available (e.g., **gp3, io2, st1, sc1**) for various workloads — from general-purpose to high-performance databases.

6. **Encryption:**
   EBS supports encryption at rest and in transit to keep data secure.

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**Uses of Amazon EBS:**

1. **Root Volume for EC2 Instances:**
   It stores the operating system and boot files of EC2 instances.
2. **Database Storage:**
   Used to store databases such as MySQL, Oracle, or PostgreSQL that require fast read/write performance.
3. **Application Data:**
   Useful for applications needing low-latency access to data, like enterprise apps or analytics tools.
4. **Backup and Recovery:**
   Snapshots allow quick backup and recovery of entire volumes.
5. **Big Data & Analytics:**
   Used to store temporary or permanent data for processing large datasets in AWS.

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**Example:**

When you launch an EC2 instance to run a website, you can attach an EBS volume to:

- Store the operating system,
- Keep website files,
- Save database data,
  and even take **snapshots** to back up the entire environment.

# Q2. Perform the following steps to create and attach an EBS volume:

a) Launch an EC2 instance if not already available.

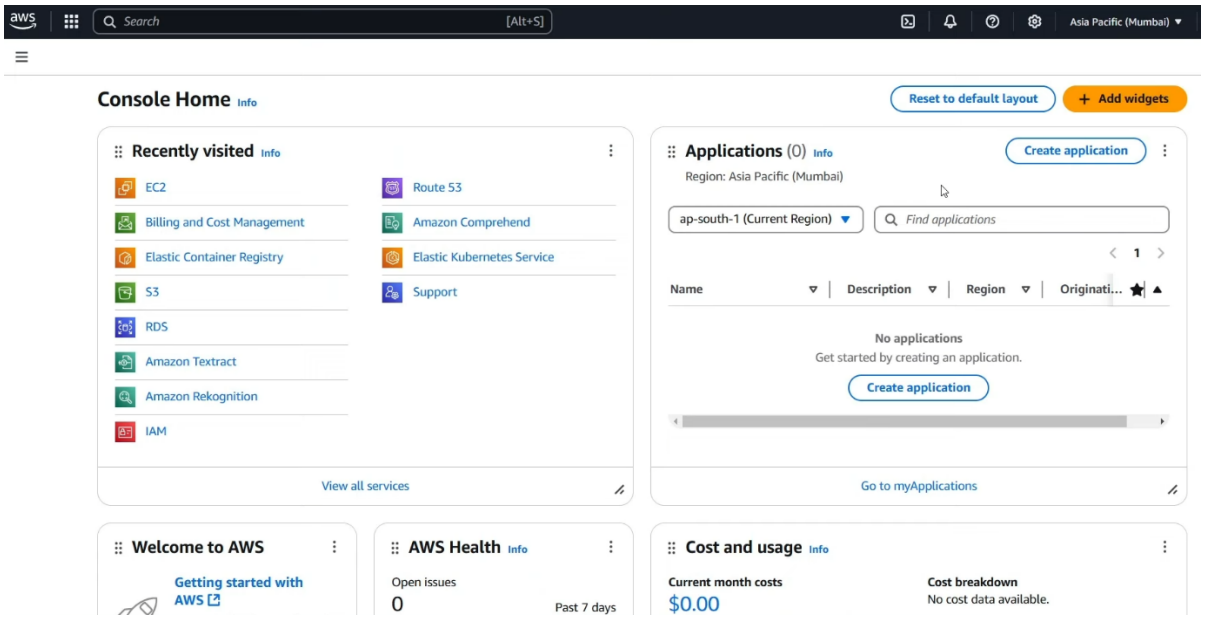
b) Create a new EBS volume in the same availability zone as the instance.

c) Attach the volume to the running EC2 instance.

d) Connect to the instance and mount the volume.

e) Format the volume and verify successful attachment.

f) Detach and delete the volume after verification.



EC2

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Compute

# Amazon Elastic Compute Cloud (EC2)

Create, manage, and monitor virtual servers in the cloud.

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.

**Launch a virtual server**

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

View dashboard

## Benefits and features

**EC2 offers ultimate scalability and control**

Fully resizable compute capacity to support virtually any workload. This service is best if you want:

- Highest level of control of the entire technology stack, allowing full integration with all AWS services
- Widest variety of server size options
- Widest availability of operating systems to choose from including Linux, Windows, and macOS
- Global scalability

**Get started**

Take our walkthroughs to help you launch an instance, learn about EC2 best practices, and set up your account.

Get started walkthroughs

Get started tutorial

EC2

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Lifecycle Manager

## Instances Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states | 1

| Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Public IPv4 |
|---|---|---|---|---|---|---|---|

No instances

You do not have any instances in this region

Launch instances

Select an instance


## Instances Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states | 1

| Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Public IPv4 |
|---|---|---|---|---|---|---|---|

No instances

You do not have any instances in this region

Launch instances

EC2 > Instances > Launch an instance

It seeems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Do not show me this message again | Take a walkthrough

## Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

### Name and tags Info

Name

e.g. My Web Server

Add additional tags

### Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and

### Summary

Number of instances Info

1

Software Image (AMI)

-

Virtual server type (instance type)

t2.micro

Firewall (security group)

-

Storage (volumes)

-

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage,

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

**Amazon Machine Image (AMI)**

Amazon Linux 2023 AMI — Free tier eligible

ami-0c50b6f7dc3701ddd (64-bit (x86), uefi-preferred) / ami-098e9400eae2749f1 (64-bit (Arm), uefi)

Virtualization: hvm  ENA enabled: true  Root device type: ebs

**Description**

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.6.20250203.1 x86_64 HVM kernel-6.1

| Architecture | Boot mode | AMI ID | Username | |
|---|---|---|---|---|
| 64-bit (x86) | uefi-preferred | ami-0c50b6f7dc3701ddd | ec2-user | Verified provider |

## Create key pair

### Key pair name

Key pairs allow you to connect to your instance securely.

key-instance

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

### Key pair type

- (●) **RSA**
  RSA encrypted private and public key pair
- ( ) **ED25519**
  ED25519 encrypted private and public key pair

### Private key file format

- (●) **.pem**
  For use with OpenSSH
- ( ) **.ppk**
  For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** Learn more

Cancel | Create key pair

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Auto-assign public IP | Info

Enable

Additional charges apply when outside of free tier allowance

**Firewall (security groups)** | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (●) Create security group
- ( ) Select existing security group

We'll create a new security group called '**launch-wizard-3**' with the following rules:

- [x] Allow SSH traffic from — Anywhere 0.0.0.0/0
  Helps you connect to your instance
- [x] Allow HTTPS traffic from the internet
  To set up an endpoint, for example when creating a web server
- [x] Allow HTTP traffic from the internet
  To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

**Configure storage** Info — Advanced

1x 8 GiB gp3 Root volume 3000 IOPS (Not encrypted)

ⓘ Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

**Summary**

Number of instances | Info

1

**Software Image (AMI)**
Canonical, Ubuntu, 24.04, amd6...read more
ami-00bb6a80f01f03502

**Virtual server type (instance type)**
t2.micro

**Firewall (security group)**
New security group

**Storage (volumes)**
1 volume(s) - 8 GiB

ⓘ **Free tier:** In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4

Cancel | Launch instance | Preview code

## Configure storage Info

Advanced

1x 8 GiB gp3 Root volume 3000 IOPS (Not encrypted)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-003c4830ec3f76355)

Launch log

### Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

1 2 3 4 5 6

**Create billing and free tier usage alerts**

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

**Connect to your instance**

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

**Connect an RDS database**

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

**Create EBS snapshot policy**

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

**Manage detailed monitoring**

**Create Load Balancer**

**Create AWS budget**

**Manage CloudWatch alarms**

New EC2 Experience

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Launch Templates
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Capacity Reservations
Images
AMIs New

Find instance by attribute or tag (case-sensitive)

| Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone |
|---|---|---|---|---|---|---|

Instance: i-07fc9b0605a1d29c0 (My First Instance)

Filter block devices

| Volume ID | Device name | Volume size (GiB) | Attachment status | Attachment time | Encrypted | K |
|---|---|---|---|---|---|---|
| vol-029f471d5b4c327f1 | /dev/xvda | 8 | Attaching | Tue Nov 15 2022 14:51:59 G... | No | – |

Recent root volume replacement tasks

Filter tasks

Replace root volume

| Task ID | Task state | Start time | Completion time | Tags |
|---|---|---|---|---|

No recent replace root volume tasks

Instance: i-07fc9b0605a1d29c0 (My First Instance)

Filter block devices

| iB) | Attachment status | Attachment time | Encrypted | KMS key ID | Delete on termination |
|---|---|---|---|---|---|
| | Attaching | Tue Nov 15 2022 14:51:59 G... | No | – | Yes |

Recent root volume replacement tasks

**Volumes** (1/1)

Actions ▼ | Create volume

Search

Volume ID = vol-029f471d5b4c327f1 | Clear filters

| IOPS | Throughput | Snapshot | Created | Availability Zone | Volume state | Alarm status |
|---|---|---|---|---|---|---|
| 100 | - | snap-08ca0f1... | 2022/11/15 14:51 GMT+5:... | ap-south-1b | In-use | No alarms |

EC2 > Volumes > Create volume

# Create volume Info

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

## Volume settings

Volume type Info

General Purpose SSD (gp2)

Size (GiB) Info

100

Min: 1 GiB, Max: 16384 GiB. The value must be an integer.

IOPS Info

300 / 3000

Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) Info

Not applicable

Availability Zone Info

ap-south-1a

Snapshot ID - *optional* Info

Don't create volume from a snapshot

Encryption Info

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

## Tags - *optional* Info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add tag

You can add 50 more tags.

Cancel | Create volume

Successfully created volume vol-00af489683ce7d086.

You can now create Amazon Data Lifecycle Manager policies to automate snapshot management directly from this screen. Select the volumes to then choose **Actions, Create snapshot lifecycle policy**. For more information, see the Knowledge Center article.

**Volumes** (1/2)

Actions

Search

| Name | Volume ID | Type | Size | IOPS | Throughput | Snapshot |
|---|---|---|---|---|---|---|
| – | vol-029f471d5b4c327f1 | gp2 | 8 GiB | 100 | - | snap-08ca0 |
| – | vol-00af489683ce7d086 | gp2 | 2 GiB | 100 | - | - |

Successfully created volume vol-00af489683ce7d086.

You can now create Amazon Data Lifecycle Manager policies to automate snapshot management directly from this screen. Select the volumes to back up, and then choose **Actions, Create snapshot lifecycle policy**. For more information, see the Knowledge Center article.

**Volumes** (1/2)

Actions

Create volume

Search

| Name | Volume ID | Type | Size | IOPS | Created |
|---|---|---|---|---|---|
| – | vol-029f471d5b4c327f1 | gp2 | 8 GiB | 100 | 2022/11/15 14 |
| – | vol-00af489683ce7d086 | gp2 | 2 GiB | 100 | 2022/11/15 15 |

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Manage tags

EC2 > Volumes > vol-00af489683ce7d086 > Attach volume

# Attach volume Info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

## Basic details

Volume ID

vol-00af489683ce7d086

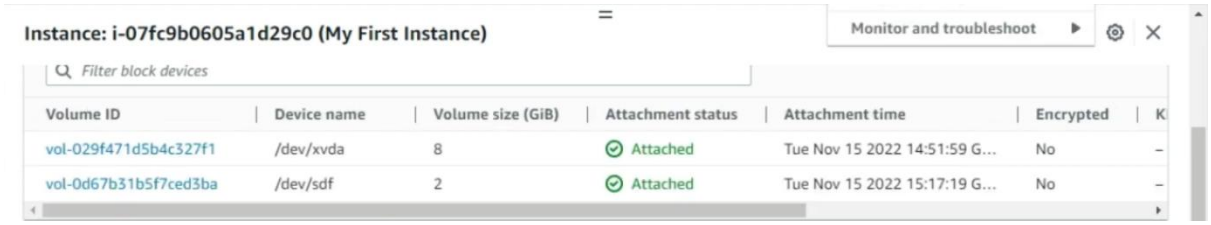
Availability Zone

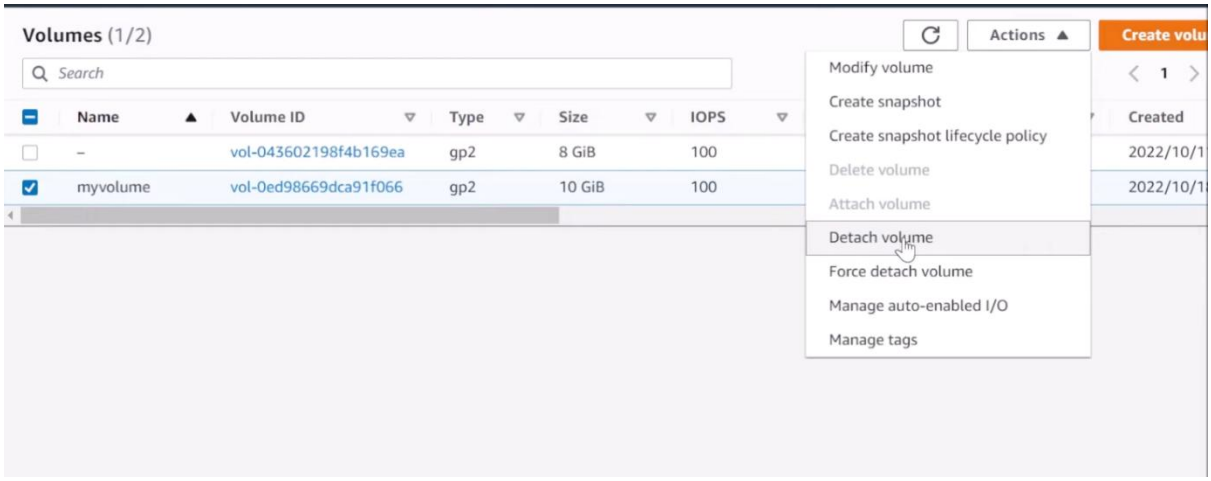
ap-south-1a

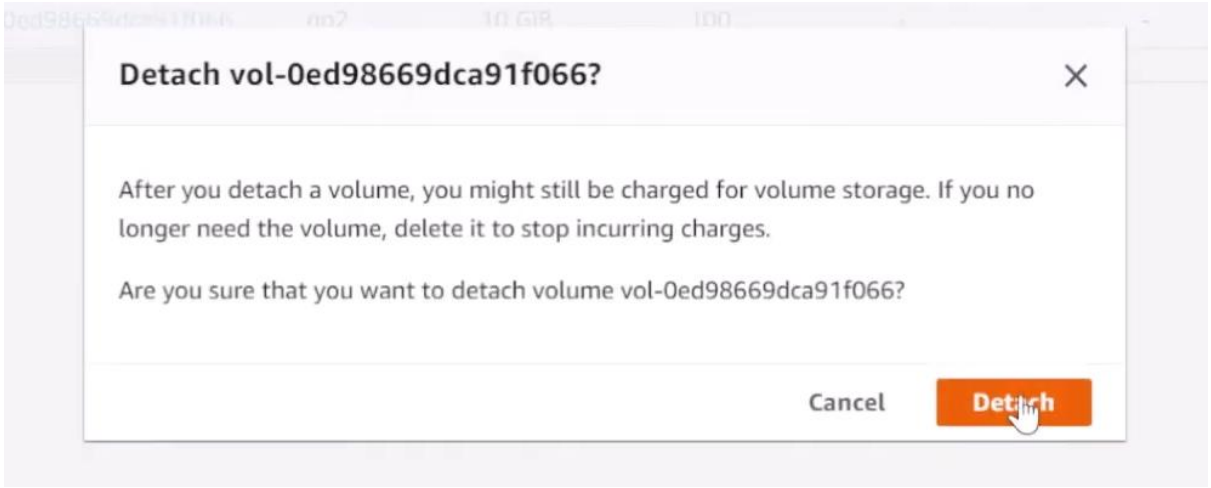
Instance Info

Only instances in the same Availability Zone as the selected volume are displayed.

Device name Info


Instance: i-07fc9b0605a1d29c0 (My First Instance)
Monitor and troubleshoot
Filter block devices
Volume ID
Device name
Volume size (GiB)
Attachment status
Attachment time
Encrypted
vol-029f471d5b4c327f1
/dev/xvda
8
Attached
Tue Nov 15 2022 14:51:59 G...
No
vol-0d67b31b5f7ced3ba
/dev/sdf
2
Attached
Tue Nov 15 2022 15:17:19 G...
No


Volumes (1/2)
Actions
Create volu
Search
Name
Volume ID
Type
Size
IOPS
Created
–
vol-043602198f4b169ea
gp2
8 GiB
100
2022/10/1
myvolume
vol-0ed98669dca91f066
gp2
10 GiB
100
2022/10/1
Modify volume
Create snapshot
Create snapshot lifecycle policy
Delete volume
Attach volume
Detach volume
Force detach volume
Manage auto-enabled I/O
Manage tags


Detach vol-0ed98669dca91f066?
After you detach a volume, you might still be charged for volume storage. If you no longer need the volume, delete it to stop incurring charges.
Are you sure that you want to detach volume vol-0ed98669dca91f066?
Cancel
Detach